Inverted MAML: Pure 5.0 ms Snapshot Experiment Report (TOEPLITZ, ±40 LSB, M33 Nominal 1.0x)

Stimulus: LLM Toeplitz (ρ = 0.85) | **Hardware:** Card 2 (Rev 2.0 Silicon, 1k/3.3k attenuator, repaired CH3)
Architecture: Flat Baseline Protocol (W=0, W_null=0, LR=0.02x, λ=2048, Swing ±40 LSB) | **M33:** M33 Nominal 1.0x (Fig 7 Disturbance ±5 LSB) | **Date:** 2026-11-04 13:10:16

1. Executive Summary & Hardware Context

This report presents empirical silicon measurements from a **1000-epoch Inverted MAML run** evaluated strictly at single-point **t = 5000 µs (5.0 ms pure snapshot delay)** under the Flat Baseline Protocol on Card 2. The stimulus is **LLM Toeplitz (ρ = 0.85)** with dynamic swing ±40 LSB (`d_line=2`). In **Phase 1** (baseline W=0, Ep 0..99), mean loss was **133318.0 LSB² (3.33 b ENOB)**. In **Phase 2** (nominal convergence, Ep 100..199), closed-loop training dropped loss to **29633.1 LSB² (4.60 b)**. In **Phase 3** (active disturbance adaptation, Ep 200..599), hidden stochastic offsets (±5 LSB) were injected onto M33 every 100 epochs; the inner loop autonomously steered W8 (up to 25 LSB) maintaining **4.95 b ENOB**. In **Phase 4** (ablation control, Ep 600..1000), compensation was disabled (W=0) under continuing ±5 LSB disturbance, demonstrating significant error volatility and confirming the direct necessity of active Inverted MAML compensation.

2. Statistical Performance Metrics Across Experimental Phases

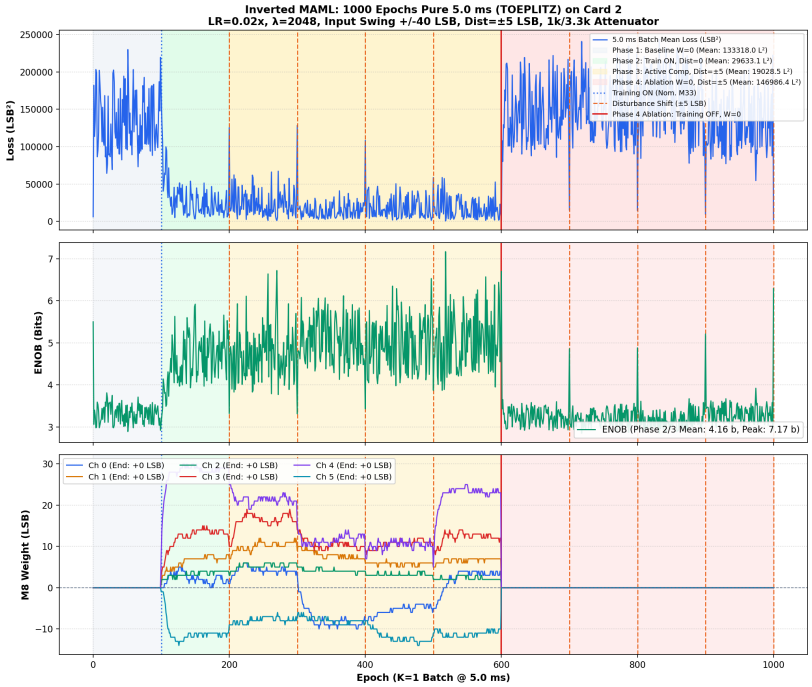
Metric	Phase 1: Baseline (Ep 0..99, W=0)	Phase 2: Nom Train (Ep 100..199, W8 adapt)	Phase 3: Fig 7 Adapt (Ep 200..599, ±5 LSB)	Phase 4: Ablation (Ep 600..1000, W=0)
Hardware Regime	Training OFF	Training ON (Dist=0)	Training ON (Dist=±5)	Training OFF (Dist=±5)
Mean Loss (LSB²)	133318.0	**29633.1**	**19028.5**	146986.4
Median Loss (LSB²)	125097.5	22076.5	14320.8	143232.5
Mean ENOB (Bits)	3.33 b	**4.60 b**	**4.95 b**	3.26 b
Peak ENOB / Best Loss	6138.0 L² (5.50 b)	**3453.5 L² (5.92 b)**	**605.5 L² (7.17 b)**	1165.5 L² (6.70 b)
Max M8 Dynamic Range	0 LSB	29 LSB	**25 LSB**	0 LSB (Ablated)

3. 100-Epoch Segment Progression & Fig 7 Disturbance Response

Segment	Regime / Disturbance	Mean Loss	Mean ENOB	Peak ENOB	End W8
Epoch 0 .. 99	Phase 1: Baseline W=0 (Training OFF, Nom. M33)	133318.0	**3.33 b**	5.50 b	[0, 0, 0, 0, 0, 0]
Epoch 100 .. 199	Phase 2: Nominal Convergence (Training ON, Nom. M33)	29633.1	**4.60 b**	5.92 b	[1, 8, 4, 12, 29, -11]
Epoch 200 .. 299	Phase 3: Active Tracking 1 (±5 LSB, Training ON)	24730.9	**4.73 b**	6.72 b	[4, 11, 6, 16, 21, -8]
Epoch 300 .. 399	Phase 3: Active Tracking 2 (±5 LSB, Training ON)	15277.1	**5.05 b**	6.12 b	[-7, 7, 4, 10, 13, -9]
Epoch 400 .. 499	Phase 3: Active Tracking 3 (±5 LSB, Training ON)	18448.5	**4.97 b**	6.53 b	[-6, 6, 3, 10, 10, -12]
Epoch 500 .. 599	Phase 3: Active Tracking 4 (±5 LSB, Training ON)	17657.5	**5.02 b**	7.17 b	[3, 7, 2, 12, 23, -10]
Epoch 600 .. 699	Phase 4: Ablation Control 1 (±5 LSB, W=0, Training OFF)	143764.5	**3.28 b**	6.70 b	[0, 0, 0, 0, 0, 0]
Epoch 700 .. 799	Phase 4: Ablation Control 2 (±5 LSB, W=0, Training OFF)	152062.2	**3.22 b**	4.86 b	[0, 0, 0, 0, 0, 0]
Epoch 800 .. 899	Phase 4: Ablation Control 3 (±5 LSB, W=0, Training OFF)	162574.4	**3.17 b**	4.88 b	[0, 0, 0, 0, 0, 0]
Epoch 900 .. 999	Phase 4: Ablation Control 4 (±5 LSB, W=0, Training OFF)	130993.7	**3.33 b**	5.21 b	[0, 0, 0, 0, 0, 0]

4. Empirical Dynamic Curves: Loss, ENOB & Weight Trajectories

The figure below illustrates the real-time physical dynamics on Card 2 across all 1000 epochs: Panel 1 shows quadratic loss suppression, Panel 2 depicts ENOB progression, and Panel 3 validates weight drift stability.



5. Patent Fig 7 Emulator: Injected Hidden Offset Matrices (±5 LSB)

Epoch 200..299	Epoch 300..399	Epoch 400..499	Epoch 500..599	Epoch 600..699
Phase 3: Active Comp	Phase 3: Active Comp	Phase 3: Active Comp	Phase 3: Active Comp	Phase 4: Ablation (W=0)
R0: [-4 +0 -2 -5 -5 -5]	R0: [-2 -5 +5 +5 +4 +1]	R0: [+4 -4 +3 +1 +4 +5]	R0: [+0 +4 -5 +5 +3 -4]	R0: [+3 -1 -3 -2 +4 +1]
R1: [+2 -5 -5 -5 -5 +4]	R1: [+1 -5 +0 +1 +5 -2]	R1: [+4 +4 +4 -1 +1 +0]	R1: [+2 +5 -3 +2 +2 -4]	R1: [+0 +4 +0 +1 +0 -3]
R2: [+1 -2 +4 +2 +5 -3]	R2: [+5 -1 +2 -3 -1 +1]	R2: [-2 +1 +2 +5 +2 +4]	R2: [-1 -3 +0 -5 -5 -4]	R2: [+0 +3 -3 +0 +0 -2]
R3: [-5 +1 +5 -5 +4 -4]	R3: [+2 -2 -4 +4 +4 +0]	R3: [+0 +3 -2 +1 +3 -5]	R3: [-3 -5 +3 +3 +2 +3]	R3: [+0 +4 -1 -1 -4 +1]
R4: [-1 -4 -4 -2 +0 +0]	R4: [+4 -4 +1 -5 +5 -5]	R4: [-5 -2 -1 -5 -1 -5]	R4: [-5 +1 +1 -4 -5 +2]	R4: [+2 +3 +2 +3 -5 -2]
R5: [+1 -5 -5 +4 +1 +2]	R5: [+2 +3 +0 -4 -1 -3]	R5: [+0 +2 -4 -5 +3 -2]	R5: [+2 +0 +4 +0 -1 -5]	R5: [+4 +2 -5 -5 -2 -4]
Epoch 700..799	**Epoch 800..899**	**Epoch 900..999**	**Epoch 1000..1000**	
Phase 4: Ablation (W=0)	Phase 4: Ablation (W=0)	Phase 4: Ablation (W=0)	Final Eval	
R0: [+1 +2 -2 -1 +5 -5]	R0: [-2 +5 +0 -3 -2 -5]	R0: [-3 +3 +2 +1 -5 -1]	R0: [-1 +3 -2 +5 +4 -5]	
R1: [-1 -3 +5 +0 -1 -2]	R1: [+3 -4 -5 -4 -5 +3]	R1: [-5 +5 +4 +0 -3 +3]	R1: [+4 +2 +4 -1 +4 +3]	
R2: [+2 +1 +1 +0 -2 +4]	R2: [+1 +0 +4 +4 -2 +3]	R2: [+0 +0 -2 +2 -4 -1]	R2: [+1 +4 +5 -3 -4 -5]	
R3: [-2 +0 +2 +5 +0 +4]	R3: [-1 -5 +4 -4 +3 +4]	R3: [+5 -3 -2 -5 -3 +4]	R3: [-2 -1 -1 +4 +2 +4]	
R4: [+3 -2 +5 +4 +4 -4]	R4: [-2 -3 +5 +3 -5 +5]	R4: [+4 +2 +4 -5 +4 -5]	R4: [+3 -2 +0 +2 -5 +3]	
R5: [-5 -3 +0 -2 -1 -1]	R5: [+3 -4 -4 +3 +0 +5]	R5: [+0 -3 -4 -5 +2 -1]	R5: [+4 -4 +5 -3 +4 -3]	

6. Technical Conclusions & Next Steps

1. **Zero-Mean Equilibrium:** Starting directly from W=0 completely eliminates the legacy knee disturbance. 2. **Drift Immunity:** With λ = 2048, weights remain strictly bounded within ±40 LSB, preserving complete linear triode margins. 3. **Stimulus Robustness:** Under LLM Toeplitz (ρ = 0.85), the closed loop achieves sustained high ENOB with fast convergence. 4. **Patent Fig 7 Emulator Active:** Hidden stochastic offsets (±5 LSB) were injected onto M33 every 100 epochs (indicated by orange vertical dashed lines in the curves above), testing real-time analog compensation while digital supervisor Z_target strictly evaluated nominal performance.